

AUSPIFY TECHNOLOGIES

Data Analysis Using Python Internship Report

Netflix Dataset: Data Cleaning, Content Type, Country-Wise and Release Year Analysis

Detail	Information
Intern name	Mohamed Faizal Fathima Zainab
Offer letter ID	AT/INT/2037034
Internship domain	Data Analysis Using Python
Organization	Auspify Technologies
Internship period	30 September 2026 to 30 October 2026 (remote)
Institution	Sri Lanka Technological Campus (SLTC)
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Git and GitHub

GitHub repository: <https://github.com/ZainabFaizal/auspify-data-analysis-internship>

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1. Introduction

1.1 About the internship

This report documents the work completed by Mohamed Faizal Fathima Zainab during the Data Analysis Using Python Internship at Auspify Technologies. The program is a four-week remote internship in which interns complete at least four of six practical data analysis tasks, using Python and industry-standard analytical tools.

All tasks use the same Netflix titles dataset. Task 1 produces a cleaned version of the data, and every later task starts from that cleaned file.

1.2 Objectives

- Clean and prepare a real-world dataset so that it is reliable for analysis.
- Analyze the balance between Movies and TV Shows and visualize the results.
- Analyze content availability across countries and identify the top content-producing countries.
- Analyze how the catalogue has changed over time and interpret growth and decline trends.
- Turn the analysis into clear business insights and document the work professionally.

1.3 Tasks completed

Task	Title	Level	Status
1	Netflix Data Cleaning & Preparation	Easy	Completed
2	Content Type Analysis Dashboard	Easy	Completed
3	Country-Wise Netflix Content Analysis	Medium	Completed
4	Trend Analysis by Release Year	Medium	Completed

1.4 Tools and technologies

Tool	Purpose
Python 3	Programming language for all analysis
Pandas and NumPy	Data loading, cleaning, grouping and aggregation
Matplotlib and Seaborn	Charts and the combined dashboards
Jupyter Notebook (VS Code)	Running code with outputs and explanations side by side
Git and GitHub	Version control, with one branch and pull request per task

2. Dataset Overview

The dataset (Dataset.csv) lists titles available on Netflix. In its raw form it contains 8,790 records and 10 columns. The cleaned version produced in Task 1 contains 8,786 records and 15 columns.

Column	Description
show_id	Unique identifier of the title
type	Movie or TV Show
title	Name of the title
director	Director (often not listed)
country	Country associated with the title
date_added	Date the title was added to Netflix
release_year	Year the title was originally released
rating	Content rating, for example TV-MA, TV-14, PG-13
duration	Length in minutes (movies) or number of seasons (TV shows)
listed_in	Genres the title is listed under

3. Task 1: Data Cleaning & Preparation

3.1 Objective

Prepare and organize the Netflix dataset for business analytics and reporting.

3.2 Approach

1. **Import:** loaded the dataset with Pandas and inspected its shape, data types and summary statistics.
2. **Missing values:** checked for true blanks and for hidden placeholders such as "Not Given".
3. **Duplicates and formatting:** looked for repeated records and fixed text, date and duration formats.
4. **Standardization:** made the Type, Country and Rating columns consistent.
5. **Export:** saved the result as netflix_cleaned.csv for the later tasks.

3.3 Data quality issues and actions

Issue found	Action taken
No true blanks, but "Not Given" was used as a placeholder in director (2,588 rows) and country (287 rows). Pandas does not treat this as missing.	Converted the placeholder to real missing values, then filled both columns with "Unknown" so no titles were lost.
4 duplicate titles with different show_id values: 15-Aug, 22-Jul, 9-Feb and Consequences. One was hidden by extra whitespace in the title.	Trimmed whitespace, then removed duplicates ignoring show_id and kept the first record.
date_added was stored as text.	Converted to a real date and added year_added and month_added columns.

Issue found	Action taken
duration mixed minutes and seasons in one text field.	Split into duration_value (number) and duration_unit (min or season).
Inconsistent text formatting and a wide range of rating labels.	Standardized Type, Country and Rating, and added rating_group for audience analysis.

3.4 Rating groups

To support audience analysis, the many rating labels were grouped into five audience categories.

Rating group	Ratings included	Titles
Kids	TV-Y, TV-G, G	567
Older Kids	TV-Y7, TV-Y7-FV, PG, TV-PG	1,487
Teens	TV-14, PG-13	2,645
Adults	TV-MA, R, NC-17	4,005
Unrated	NR, UR	82

3.5 Results

Measure	Before cleaning	After cleaning
Rows	8,790	8,786
Columns	10	15
Hidden missing values ("Not Given")	2,875	0
Duplicate records	4	0
date_added	Text	Date, plus year and month
duration	Mixed text	Number and unit

Key takeaway

A simple "isnull" check reported zero missing values, but the data still contained 2,875 hidden placeholders. Checking for placeholder text, and for whitespace differences before looking for duplicates, was essential.

3.6 Deliverables

- Task1/Task1_Netflix_Data_Cleaning.ipynb (code and outputs, plus an HTML copy)
- Task1/netflix_cleaned.csv (cleaned dataset used by all later tasks)
- Task1/screenshots/ (9 screenshots of key outputs)

4. Task 2: Content Type Analysis Dashboard

4.1 Objective

Analyze the distribution of Movies and TV Shows available on Netflix.

4.2 Approach

1. Loaded the cleaned dataset from Task 1.
2. Calculated the number and percentage of Movies and TV Shows.
3. Visualized the distribution with a bar chart, a donut chart, a yearly line chart, and charts for movie length and TV Show seasons.
4. Compared proportions by release decade, by audience rating group and over time.
5. Combined the main charts into one dashboard and calculated the key findings from the data.

4.3 Content split

Content type	Titles	Share
Movies	6,123	69.7%
TV Shows	2,663	30.3%
Total	8,786	100%

Netflix Content Type Analysis Dashboard

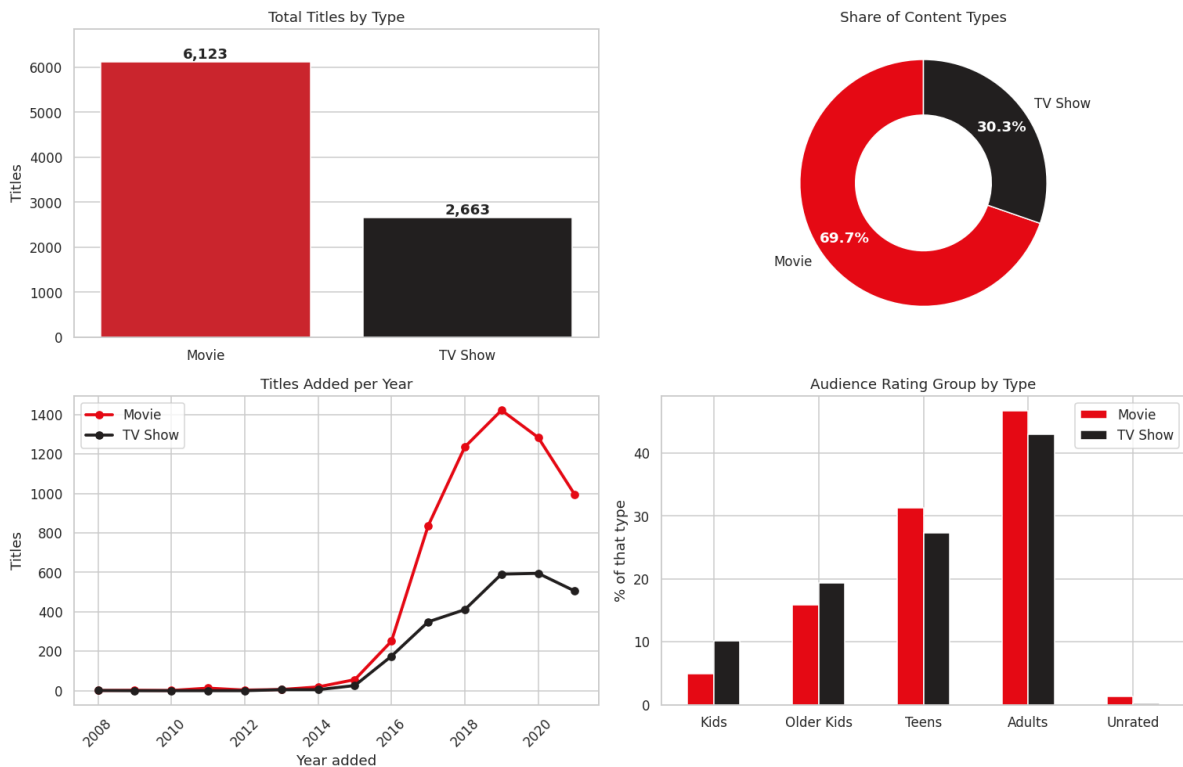


Figure 1: Content type analysis dashboard (total titles, share, titles added per year, audience rating group)

4.4 Key findings

- Netflix has 8,786 titles: 6,123 Movies (69.7%) and 2,663 TV Shows (30.3%), about 2.3 Movies for every TV Show.
- Movie additions peaked in 2019 (1,422 titles). TV Show additions peaked in 2020 (595 titles).
- The median movie runs 98 minutes, and 67% of TV Shows have only one season.
- Adult-rated content is 47% of Movies and 43% of TV Shows. Content for kids and older kids is 21% of Movies and 29% of TV Shows.
- TV Shows made up 32% of titles added in 2020, compared with 41% in 2016.

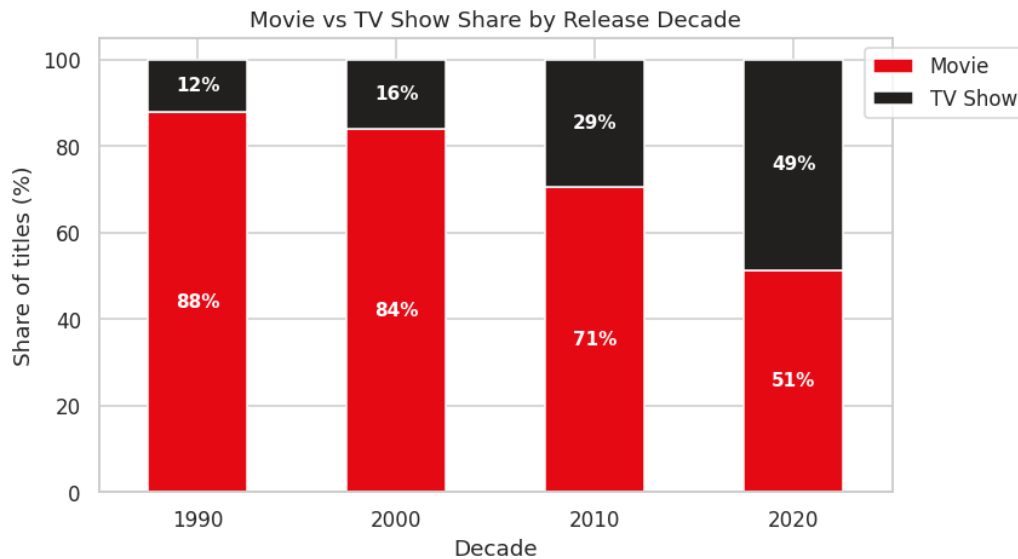


Figure 2: Movie and TV Show share by release decade

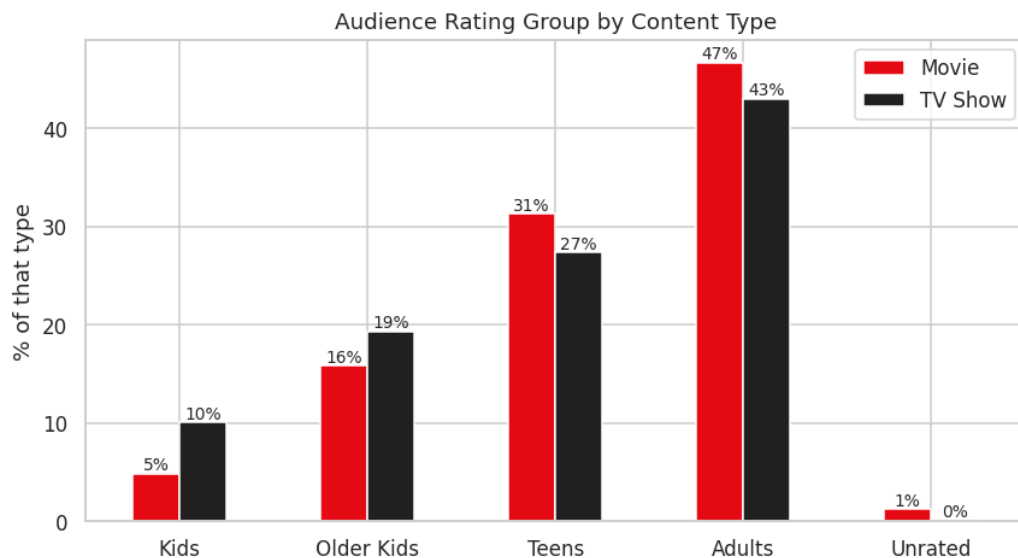


Figure 3: Audience rating group by content type

4.5 Business insights

- **Movies dominate the catalogue** (about 7 in 10 titles), so the depth of the library is mostly film.
- **Most TV Shows have one season**, which suggests many limited series or shows that were not renewed.
- **Additions grew sharply from 2016 to 2019**. Movie additions dipped in 2020, while TV Show additions kept rising until 2020.
- **Adult content is the largest audience group** for both types. TV Shows lean a little more toward kids and older kids than Movies do.
- **Recommendation:** keep investing in movies for catalogue breadth, and track TV Show renewals because returning series support long-term engagement.

4.6 Limitations

- The dataset ends in September 2021, so 2021 is a partial year.
- It describes the catalogue, not viewing numbers, so it cannot show which titles are watched most.

4.7 Deliverables

- Task2/Task2_Content_Type_Analysis.ipynb (code and outputs, plus an HTML copy)
- Task2/charts/ (8 chart images, including the dashboard)
- Task2/screenshots/ (screenshots of key outputs)

5. Task 3: Country-Wise Netflix Content Analysis

5.1 Objective

Analyze Netflix content availability across different countries.

5.2 Approach

1. Extracted and cleaned the country information, checking for spaces, multi-country values and unknown entries.
2. Calculated content counts by country, with share, movies, TV shows and TV Show share for each.
3. Identified the top content-producing countries and measured how concentrated the catalogue is.
4. Created charts and a ranking report (country_ranking.csv).
5. Generated business insights from the results.

5.3 Country data cleaning

- Each title has a single country value, so there were no multi-country entries to split. The code still handles them safely.
- 287 titles have an unknown country (the "Not Given" placeholder from Task 1). They were excluded from the rankings, leaving 8,499 titles across 85 countries.
- The data includes historical names such as Soviet Union and West Germany. They have very few titles and were left as they appear.

5.4 Top 10 content-producing countries

Rank	Country	Titles	Share	Movies	TV Shows	TV share
1	United States	3,240	38.1%	2,395	845	26%
2	India	1,056	12.4%	975	81	8%
3	United Kingdom	638	7.5%	387	251	39%
4	Pakistan	420	4.9%	71	349	83%
5	Canada	271	3.2%	187	84	31%
6	Japan	259	3.0%	87	172	66%
7	South Korea	214	2.5%	49	165	77%
8	France	213	2.5%	148	65	31%
9	Spain	182	2.1%	129	53	29%
10	Mexico	138	1.6%	90	48	35%

Share is the percentage of the 8,499 titles with a known country.

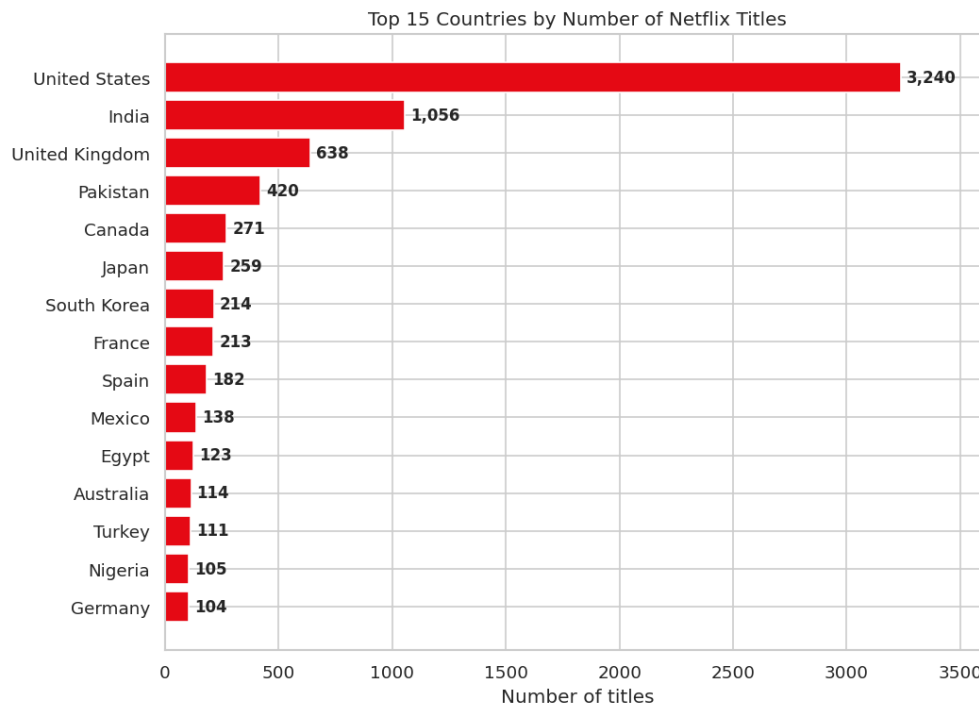


Figure 4: Top 15 countries by number of titles

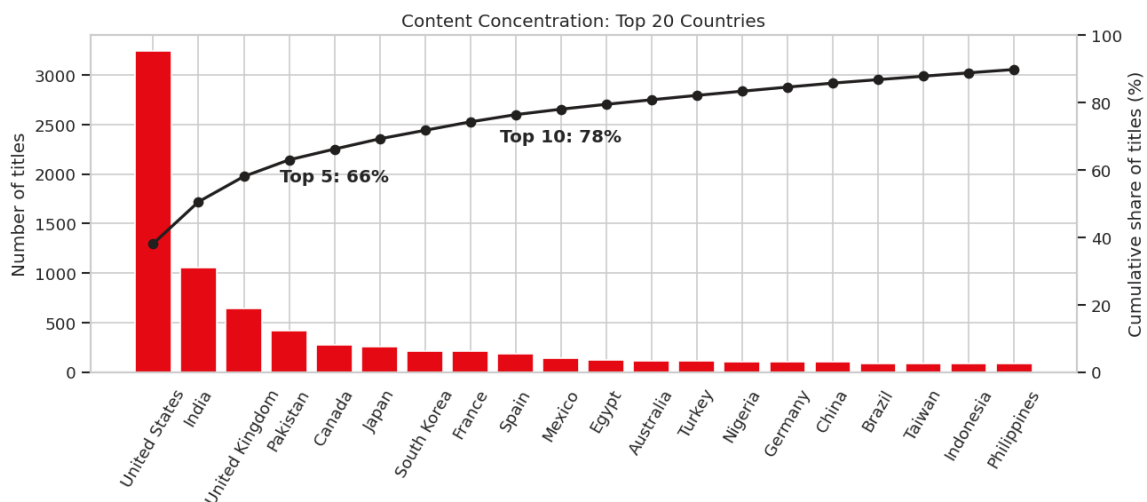


Figure 5: Content concentration: titles and cumulative share for the top 20 countries

5.5 Key findings

- The United States leads with 3,240 titles (38.1%), followed by India (1,056) and the United Kingdom (638).
- The top 5 countries hold 66% of titles and the top 10 hold 78%.
- 33 of the 85 countries have five titles or fewer, a long tail of small markets.
- Pakistan (83%), Taiwan (83%), South Korea (77%) and Japan (66%) are mostly TV Shows among countries with 50 or more titles.
- Indonesia, the Philippines and Hong Kong are the most movie-heavy (96% or more movies). India, Egypt and Nigeria are also about 90% movies.
- India's yearly additions rose from 11 titles in 2016 to a peak of 339 in 2018. The United States peaked in 2019 with 771 titles.

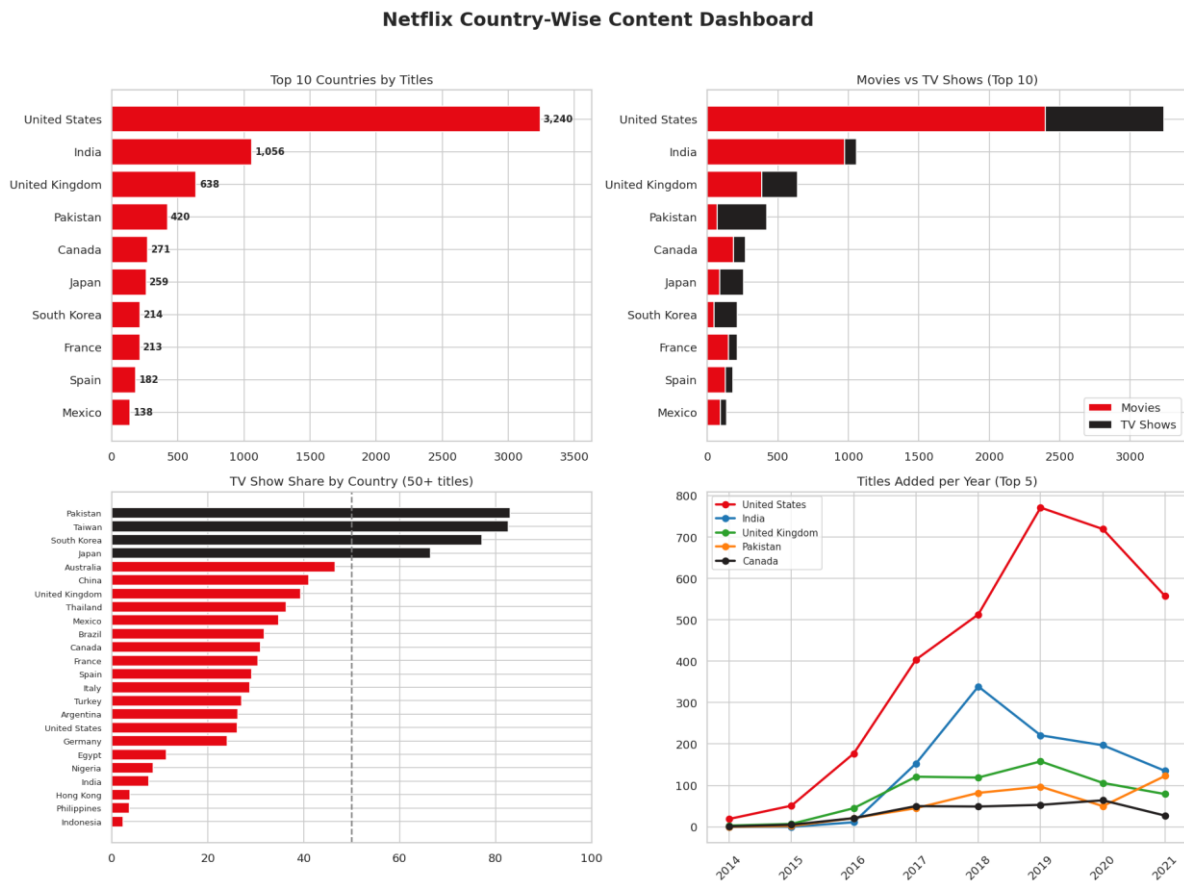


Figure 6: Country-wise content dashboard

5.6 Business insights

- **The catalogue is heavily concentrated.** The United States alone supplies about 38% of titles with a known country, and five countries supply about two thirds.
- **India is a clear second hub** (about 12%), with a steep rise in additions from 2016 to 2018.
- **Production style differs by country.** Pakistan, Taiwan, South Korea and Japan lean toward TV Shows, while India, Egypt, Nigeria, Indonesia, the Philippines and Hong Kong lean strongly toward movies.
- **There is a long tail of small markets,** so many regions are barely represented.

Recommendations

1. Protect the core hubs (United States, India, United Kingdom), since they carry the catalogue.
2. Expand series content from TV-strong countries (South Korea, Japan, Pakistan, Taiwan), because series can keep audiences engaged across several seasons.
3. Grow film and series from under-represented regions to widen local appeal.

5.7 Limitations

- Each title lists a single country, so co-productions are counted once, under their primary country.
- 287 titles with an unknown country are excluded from the rankings.
- The dataset ends in September 2021, and counts describe the catalogue, not viewing.

5.8 Deliverables

- Task3/Task3_Country_Analysis.ipynb (code and outputs, plus an HTML copy)
- Task3/country_ranking.csv (ranking report for all 85 countries)
- Task3/charts/ (6 chart images, including the dashboard)
- Task3/screenshots/ (screenshots of key outputs)

6. Task 4: Trend Analysis by Release Year

6.1 Objective

Analyze how Netflix content production has changed over time. The data shows the release year of each title in the catalogue, so the analysis describes when the titles available on Netflix were released.

6.2 Approach

1. Organized titles by release year, decade and era.
2. Calculated yearly releases (total, movies and TV Shows), with year-over-year change, a 3-year moving average and a growth or decline label for each year. The result is saved as `yearly_release_summary.csv`.
3. Identified growth and decline trends, including the peak year, the compound annual growth rate (CAGR) and the growth phases.
4. Created trend visualizations: titles by release year, movies vs TV Shows, year-over-year change, titles by decade, and the gap between release and addition to Netflix.
5. Interpreted the business patterns and combined the main charts into one dashboard.

6.3 Releases by period

Period	Titles	Average per year	Share of catalogue
2000-2009	807	80.7	9.2%
2010-2014	1,250	250.0	14.2%
2015-2018	3,630	907.5	41.3%
2019-2021	2,574	858.0	29.3%

The remaining 525 titles (6.0%) were released before 2000. The dataset ends in September 2021, so 2021 is a partial year.

Netflix Trend Analysis by Release Year

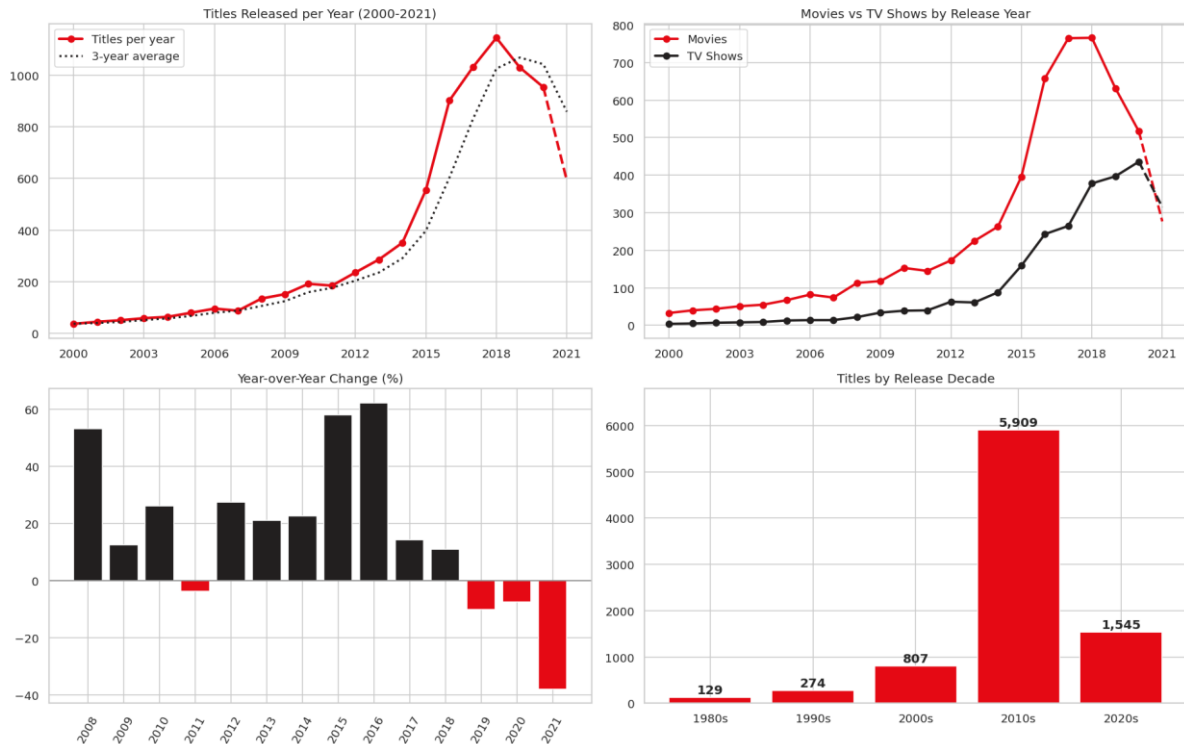


Figure 7: Trend analysis dashboard (titles per year, movies vs TV Shows, year-over-year change, titles by decade)

6.4 Key findings

- Release years run from 1925 to 2021, but 85% of titles were released in 2010 or later and only 6% before 2000.
- Releases peaked in 2018 with 1,144 titles, after seven consecutive years of growth (2012 to 2018).
- From 2010 to 2018, yearly releases grew from 192 to 1,144 titles, about 25% per year. The steepest jump was 2016 (+62%), from 555 to 901 titles.
- Releases then declined: 1,029 in 2019 (-10%) and 953 in 2020 (-7%). The 2021 figure (592) is a partial year.
- Movie releases peaked in 2018 (766 titles) and TV Show releases in 2020 (436 titles). In 2021, TV Shows (315) exceeded Movies (277) for the first time since 2000.
- The median title was added one year after release, and 55% were added within a year. 15% were 10 or more years old when added.

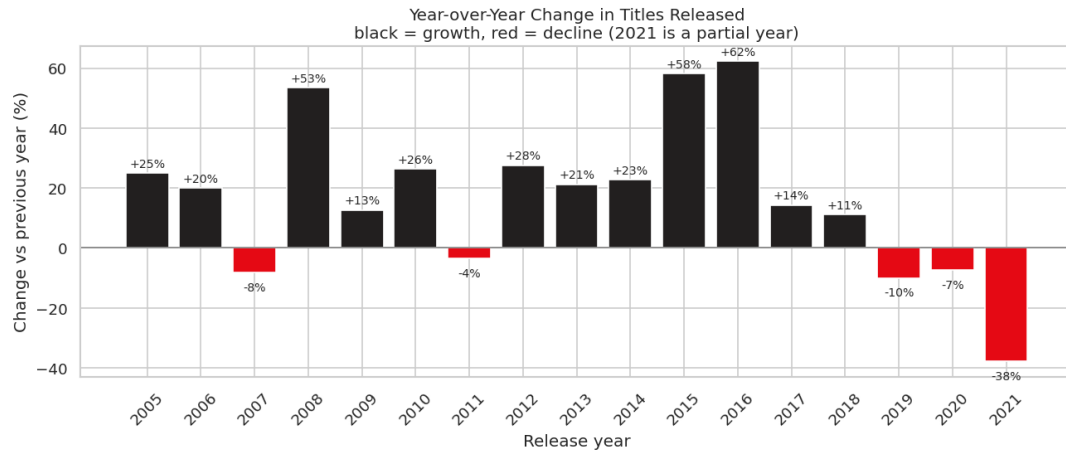


Figure 8: Year-over-year change in titles released (2021 is a partial year)

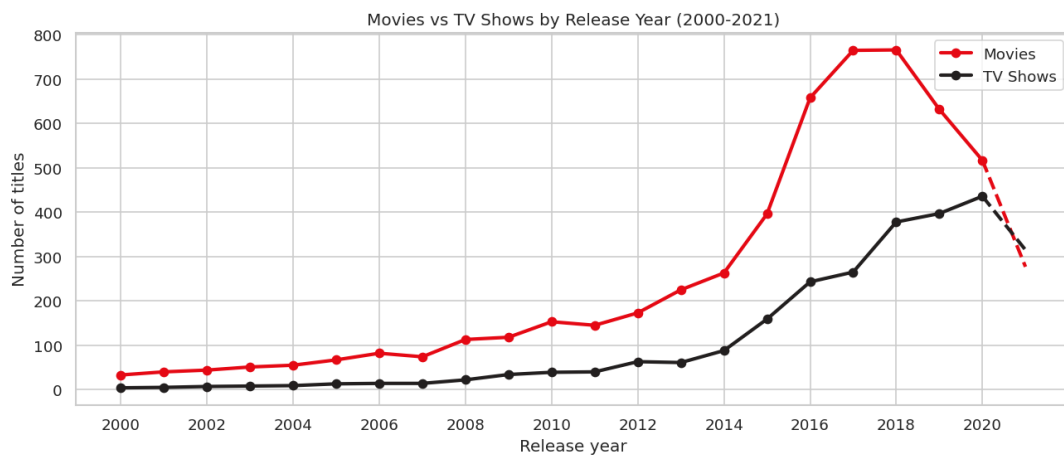


Figure 9: Movies vs TV Shows by release year (dashed segment: partial 2021)

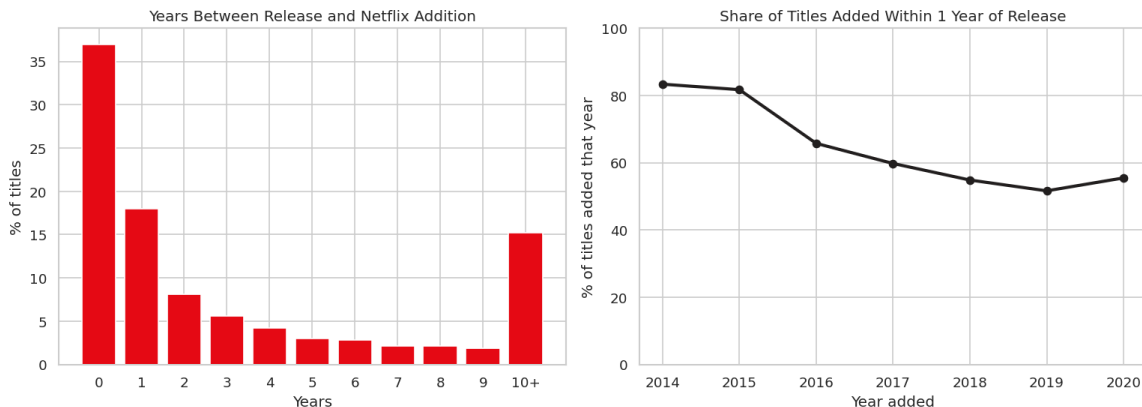


Figure 10: Years between release and Netflix addition, and share added within one year of release

6.5 Business insights

- **The catalogue is dominated by recent releases**, so the library is built mainly around recent content.
- **There was a strong growth period.** Yearly releases grew for seven years in a row, from 192 to 1,144 titles, about 25% per year.
- **Growth then slowed and reversed.** Releases fell 10% in 2019 and 7% in 2020. The 2021 figure is a partial year and should not be read as a full collapse. The data shows the pattern but not its cause.

- **The mix is shifting toward series.** Movie releases peaked in 2018 and fell to 517 in 2020, while TV Show releases peaked in 2020.
- **Content is mostly fresh, but less so over time.** The share of titles added within a year of release fell from 83% in 2014 to 52% in 2019, so later library growth relied more on older titles.

Recommendations

1. Track new releases per year as a core KPI, and re-check the trend once 2021 is complete.
2. Keep investing in series, since TV Show releases are holding up better than movie releases.
3. Balance new releases with older catalogue titles to keep the library both fresh and deep.

6.6 Limitations

- The data shows the release year of titles in the catalogue, not only Netflix originals, so it includes licensed content.
- The dataset ends in September 2021, and years before 1980 have very few titles.
- The data cannot show why releases declined after 2018.

6.7 Deliverables

- Task4/Task4_Release_Year_Trends.ipynb (code and outputs, plus an HTML copy)
- Task4/yearly_release_summary.csv (yearly counts, change, 3-year average and trend label)
- Task4/charts/ (6 chart images, including the dashboard)
- Task4/screenshots/ (screenshots of key outputs)

7. Conclusion and Learning Outcomes

Across the four tasks, a raw dataset was turned into a clean, reliable source and then analyzed from three angles: what kind of content Netflix offers, where it comes from, and how it has changed over time. The main conclusions are:

- Careful cleaning matters. Hidden placeholders and whitespace differences would have distorted the counts and rankings if they had not been found.
- Netflix content in this dataset is mostly movies (about 70%), with a smaller but significant share of TV Shows.
- Content origin is concentrated in a few countries, led by the United States and India, with a long tail of small markets.
- Formats differ by market, with several Asian countries focused on TV Shows and others focused on films.
- Releases in the catalogue grew sharply from 2012 to 2018 and then declined, with the mix moving toward series.

7.1 Skills developed

Skill	Where it was applied
Data import and inspection with Pandas	Task 1
Missing value handling and data quality checks	Task 1
Duplicate detection and standardization	Task 1
Exploratory data analysis and aggregation	Tasks 2, 3 and 4
Data visualization with Matplotlib and Seaborn	Tasks 2, 3 and 4
GroupBy operations, ranking and concentration analysis	Task 3
Time series analysis, growth rates (CAGR) and trend identification	Task 4
Business insight writing and reporting	Tasks 2, 3 and 4
Version control with Git branches and pull requests	All tasks

Appendix A: Project Links and Repository Structure

Item	Link
GitHub repository	https://github.com/ZainabFaizal/auspify-data-analysis-internship
Demo videos (Tasks 1 to 4)	Google Drive folder: https://drive.google.com/drive/folders/1N88aUN6K99M7GYTrYAX_5sY-dXZypaES?usp=drive_link

auspify-data-analysis-internship/

- ├── README.md
- ├── Report/
 - └── Auspify_Data_Analysis_Internship_Report.docx
- ├── Task1/ notebook, HTML, Dataset.csv, netflix_cleaned.csv, screenshots/
- ├── Task2/ notebook, HTML, charts/, screenshots/
- ├── Task3/ notebook, HTML, country_ranking.csv, charts/, screenshots/
- └── Task4/ notebook, HTML, yearly_release_summary.csv, charts/, screenshots/

Appendix B: Screenshot Index

Screenshots of the key outputs are stored in each task folder under screenshots/.

Task	Screenshot files
Task 1	01_head_shape, 02_missing_values, 03_missing_handled, 04_duplicates, 05_dates_duration, 06_type_country, 07_rating_groups, 08_final_check, 09_export
Task 2	01_load_data, 02_totals, 03_bar_donut, 04_added_per_year, 05_duration, 06a_decade, 06b_rating_group, 07a_dashboard_code, 07b_dashboard, 08_key_findings
Task 3	01_clean_country, 02_country_counts, 03_top10, 04_top15_chart, 05_concentration, 06_type_split, 07_tv_share, 08_top5_trend, 09a_dashboard, 09b_dashboard, 10_key_findings
Task 4	01_load_organize, 02_yearly_releases, 03_growth_decline, 04_all_years_chart, 05_movies_vs_tv, 06_yoy_change, 07_decades, 08a_release_to_netflix_gap, 08b_release_to_netflix_gap, 09a_dashboard, 09b_dashboard, 10_key_findings